# Manifold Theory

PENG

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# 1 Set Theory

## 1.1 Relations

**Definition 1.1.** *Equivalence relation* $\sim$*:*

- *reflexive:* $x \sim x$
- *symmetric:* $x \sim y \Rightarrow y \sim x$
- *transitive:* $x \sim y, y \sim z \Rightarrow x \sim z$

**Definition 1.2.** *Equivalence class of each* $x \in X$*:*
$[x] = \{y \in X : y \sim x\}$

**Definition 1.3.** *The collection of all* $[x]$ *is* $X/\sim$

**Example 1.4.** *Let* $A = \{1, 2, 3\}$*,* $R = \{(x, y) : x = y\}$ *on* $A$

- *reflexive:* $x = x$
- *symmetric:* $x = y \Rightarrow y = x$
- *transitive:* $x = y, y = z \Rightarrow x = z$

*Thus* $R$ *is an equivalence relation.*
*Equivalence class:*
$[1] = \{1\}$
$[2] = \{2\}$
$[3] = \{3\}$

**Example 1.5.** *Let* $A = \{1, 2, 3\}$*,* $R = \{(x, y) : x + y \text{ is even}\}$ *on* $A$

- *reflexive:* $x + x$ *is even*
- *symmetric:* $x + y$ *is even* $\Rightarrow y + x$ *is even*
- *transitive:* $x + y$ *is even,* $y + z$ *is even* $\Rightarrow \begin{cases} x \text{ is even}, y \text{ is even}, z \text{ is even} \\ x \text{ is odd}, y \text{ is odd}, z \text{ is odd} \end{cases} \Rightarrow x + z$ *is even*

*Thus $R$ is an equivalence relation.*
*Equivalence class:*
$[1] = \{1, 3\}$
$[2] = \{2\}$
$[3] = \{1, 3\}$

**Definition 1.6.** *A partition of $X$ is a collection of disjoint nonempty subsets of $X$ whose union is $X$.*

**Example 1.7.** $X = \{1, 2, 3\}$
$P = \{\{2\}, \{1, 3\}\}$
$P = \{\{1\}, \{3\}, \{2\}\}$
$P = \{\{1, 2, 3\}\}$
$\{\{1\}, \{2\}, \{\}\}$*: nonempty*
$\{\{1\}, \{2\}\}$*: union is not $X$*
$\{\{1, 3\}, \{2, 3\}\}$*: disjoint*
$\{\{1, 3\}, \{2, 4\}\}$*: union is not $X$*

**Exercise 1.8.** *Collection of equivalence class $\Rightarrow$ partition*
*Partition $\Rightarrow$ unique equivalence relation*

*Proof.* $X/\sim \Rightarrow$ partition:
For $\forall x_1, x_2 \in X$ :
$[x_1] = \{y \in X : y \sim x_1\}$
$[x_2] = \{y \in X : y \sim x_2\}$
If $x_1 \nsim x_2, \forall y \in [x_1] \Rightarrow y \notin [x_2] \Rightarrow$ disjoint nonempty subsets
If $x_1 \sim x_2, \forall y \in [x_1] \Rightarrow y \in [x_2] \Rightarrow [x_1] = [x_2]$ □

*Proof.* Partition $\Rightarrow\sim$ □

# 2 Euclidean Spaces

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

**Definition 2.1.** *For vectors:*

$$x \cdot y = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

**Definition 2.2.** *Norm of $x$:*

$$|x| = \sqrt{(x_1)^2 + (x_2)^2 + \cdots + (x_n)^2}$$

**Exercise 2.3.**

$$\max\{|x_1|, |x_2|, \ldots, |x_n|\} \leq |x| \leq \sqrt{n} \max\{|x_1|, |x_2|, \ldots, |x_n|\}$$

*Proof.* Let $x_m = \max\{|x_1|, |x_2|, \ldots, |x_n|\}$:

$$\sqrt{(x_m)^2} \leq \sqrt{(x_1)^2 + \cdots + (x_m)^2 + \cdots + (x_n)^2} \leq \sqrt{(x_m)^2 + \cdots + (x_m)^2}$$

□

**Definition 2.4.** *$\theta$ between $x$ and $y$:*

# 3 Topological Spaces

Topology [1]
Four mechanisms (normal, segment, merge, torus) exist in every fluid point: fractal.

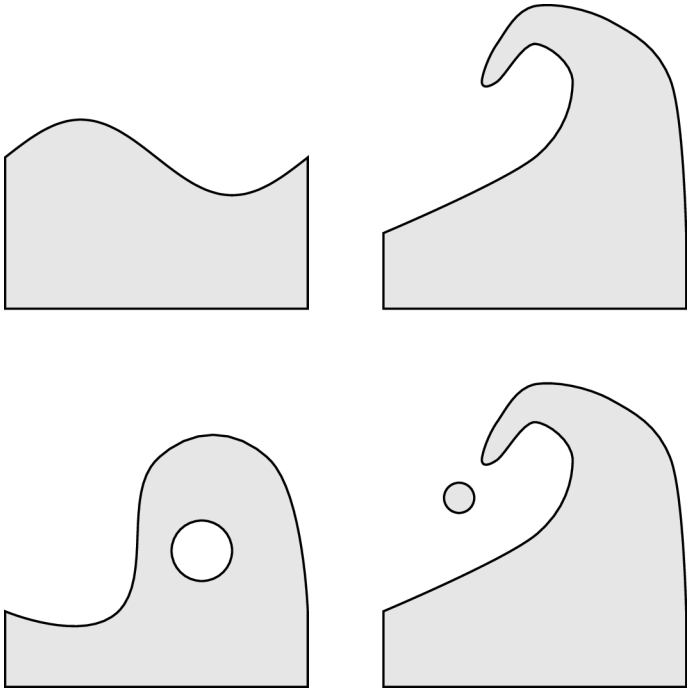


Figure 3.1: Sloshing topology.

## References

[1] J.M. Lee. *Introduction to Topological Manifolds*. Springer New York, 2010. ISBN: 9781441979391.